3_min_max_sum_avg

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#include <iostream> // For input and output (cout, cin)
#include <omp.h> // For OpenMP parallel programming
#include <climits> // For INT_MAX and INT_MIN constants
using namespace std;
// Function to find minimum value using parallel reduction
void min_reduction(int arr[], int n)
{
int min_value = INT_MAX; // Initialize min_value with largest possible integer
// OpenMP parallel loop with reduction (min operation)
#pragma omp parallel for reduction(min: min_value)
for (int i = 0; i < n; i++)
{
// Compare each element with current min_value
if (arr[i] < min_value)
{
min_value = arr[i]; // Update min_value if smaller element found
}
}
// Print minimum value
cout << "Minimum value: " << min_value << endl;
}
// Function to find maximum value using parallel reduction
void max_reduction(int arr[], int n)
{
int max_value = INT_MIN; // Initialize max_value with smallest possible integer
// OpenMP parallel loop with reduction (max operation)
#pragma omp parallel for reduction(max: max_value)
for (int i = 0; i < n; i++)
{
// Compare each element with current max_value
if (arr[i] > max_value)
{
max_value = arr[i]; // Update max_value if larger element found
}

}
// Print maximum value
cout << "Maximum value: " << max_value << endl;
}
// Function to calculate sum using parallel reduction
void sum_reduction(int arr[], int n)
{
int sum = 0; // Initialize sum to 0
// OpenMP parallel loop with reduction (addition)
#pragma omp parallel for reduction(+: sum)
for (int i = 0; i < n; i++)
```

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{
sum += arr[i]; // Add each element to sum
}
// Print total sum
cout << "Sum: " << sum << endl;
}
// Function to calculate average using parallel reduction
void average_reduction(int arr[], int n)
{
int sum = 0; // Initialize sum to 0
// OpenMP parallel loop with reduction (addition)
#pragma omp parallel for reduction(+: sum)
for (int i = 0; i < n; i++)
{
sum += arr[i]; // Add each element
}
// Calculate and print average
// Note: should be sum/n, not (n-1)
cout << "Average: " << (double)sum / n << endl;
}
// Main function
int main()
{
int *arr, n; // Pointer for array and variable for size
// Take number of elements from user

cout << "\nEnter total number of elements => ";
cin >> n;
arr = new int[n]; // Dynamically allocate array of size n
// Take array elements as input
cout << "\nEnter elements => ";
for (int i = 0; i < n; i++)
{
cin >> arr[i];
}
// Call reduction functions
min_reduction(arr, n); // Find minimum value
max_reduction(arr, n); // Find maximum value
sum_reduction(arr, n); // Find sum
average_reduction(arr, n); // Find average
return 0; // End of program
}
```